JACK J. BURLESON

AI / MACHINE LEARNING • DATA SCIENCE

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PROFESSIONAL SUMMARY

Data scientist and ML engineer completing an M.S. in Psychology: Data Science in Human Behavior at UW–Madison (Dec 2026), building AI systems end to end: multi-agent LLM document pipelines, calibrated transformer classifiers, LLM evaluation, and behavioral prediction models. Capstone with American Family Insurance on cutting AI inference cost in document processing, plus five years of high-volume logistics experience in quality assurance and process improvement.

EDUCATION

University of Wisconsin–Madison *Madison, WI* Expected Dec 2026

M.S. Candidate — Psychology: Data Science in Human Behavior · **GPA 3.85**

- Coursework in machine learning, statistics, predictive modeling, experimental design, and data visualization.

University of Wisconsin–Parkside *Kenosha, WI* Sep 2020 – May 2025

B.S., Psychology · Concentration: Neuroscience · **GPA 3.93**

- Sociology minor; Provost's List (7x), Dean's List (8x), Distinguished Achievement Award (2025), Psychological Research Certification (300 lab hours).

PROFESSIONAL EXPERIENCE

UW–Parkside — Undergraduate Researcher, Cognitive Neuroscience Lab *Kenosha, WI*

Jun 2024 – Aug 2025

- Designed and pilot-tested a study of perceived chatbot truthfulness, deception, and user cognitive workload under Dr. Melissa Gregg (code: github.com/Exios66/truth-deception-architecture).
- Operated EEG systems and analyzed neural time-series data for lab research on sleep quality and cognitive load.
- Presented findings at the 2025 Midwestern Psychological Association conference.

Avis Budget Group — Driver *Madison, WI* Jun 2026 – Present

- Move, stage, and safety-check vehicles across a 500+ vehicle rental fleet so cars are ready on time.
- Coordinate vehicle transfers between locations while upholding fleet safety standards.

UW–Madison Recreation & Wellbeing — Building Services *Madison, WI* Jan 2026 – Present

- Keep recreation and wellbeing facilities across campus clean, safe, and ready for use.
- Troubleshoot facility and equipment issues across multiple sites so centers stay safe and open during peak hours.
- Work with operations and event staff to resolve facility issues quickly and minimize disruption.

Meijer Distribution — Logistics Coordinator *Pleasant Prairie, WI* Jul 2020 – Dec 2025

- Promoted from Retail Loader to Logistics Coordinator for consistent quality assurance and process documentation.
- Analyzed quality data for 64 daily outbound trailers across the Midwest network, holding error rates below 1%.
- Partnered with warehouse and transportation teams to resolve quality issues and exceptions before shipment.

TECHNICAL SKILLS

Languages	Python, R, SQL, MATLAB
ML & Statistics	Transformer fine-tuning, probability calibration, cross-validation, SHAP, factor analysis
LLM Engineering	LangGraph, multi-agent pipelines, prompt engineering, LLM evaluation & LLM-as-a-judge
Libraries	pandas, NumPy, SciPy, scikit-learn, XGBoost, PyTorch, Hugging Face Transformers
Tools & Cloud	Git, Docker, pytest, ONNX, FastAPI, Modal, vLLM, Ollama, Parquet, AWS, Azure

PROJECTS

Digital-Mailroom *DSHB Capstone · American Family Insurance (industry partner)*

- Capstone on cutting AI inference cost in document processing: 7 Python packages covering entity extraction, classification, routing, and scalable workflows.
- Designed one LangGraph pipeline that sends each document to the cheapest step that can handle it, joining legal-document routing, prompt experiments, deterministic scoring, dataset management, and reproducible Docker deployment.

Mailroom-ML *ML Training, Evaluation & Fine-Tuning*

- Fine-tuned a ModernBERT document classifier to 95.05% document-type accuracy (307 of 323) and 0.0152 window-level expected calibration error on the test split, with per-head temperature scaling and a confidence gate that decides which documents can skip an LLM.
- Audited the training data, found a label leak (42.8% of fallback filenames named the subclass), removed it, and added a build-time guard. Trained in PyTorch on Modal GPUs, exported to ONNX for FastAPI serving, and kept 485 pytest tests in CI.

Mailroom-Corpus-EDA *Data Engineering & Dataset Audit*

- Assembled a 3,302-document, 5-class mailroom corpus from public sources (CUAD, MAUD, SEC EDGAR, Enron email, synthetic insurance claims) with stable document IDs, content hashes, per-label provenance, and tracked licenses.
- Built a 7-phase audit pipeline that reruns with one command (integrity, composition, 48 figures, 17 tables), backed by about 200 tests; all 13,753 contract clause annotations match the document text exactly.

RVL-CDIP-Classifier *Computer Vision & Prompt Engineering*

- Built a Python toolkit that classifies scanned pages into the 16 RVL-CDIP categories with OpenRouter vision models at about $0.0004 per image, plus preprocessing that converts TIFF/PDF pages to 300-DPI images and runs Tesseract OCR.
- Iterated through 29 registered prompt versions using confusion matrices, misclassification analysis, and cost projections; the v11.8 prompt scored 82.6% on a 1,120-image class-balanced evaluation slice.

Psychometric Digital Twin *Behavioral ML Research · UW–Madison*

- Predicted communication-apprehension (PRCA) scores for 241 analyzed survey respondents with Ridge, Elastic Net, k-NN, random forest, histogram gradient boosting, XGBoost, and MLP models, using cross-validation and SHAP feature importance.
- Checked the PRCA's factor structure, then benchmarked LLM "digital twin" personas (vLLM) against the ML baselines on shared metrics (MAE, band accuracy, F1, AUC); the personas did not beat the classical models, and their bias was analyzed by demographic group.

Langton's Ant *Simulation · TypeScript*

- Built a browser simulator for Langton's ant and other turmites: any rule up to 32 colors and 16 states on an unbounded, chunked grid, with a gallery of 20 published machines and shareable permalinks.
- Kept the engine deterministic (integer-only stepping, seeded randomness) and covered it with 99 passing tests, including a pinned determinism test.

RESEARCH TRAINING

CITI Program: Conflicts of Interest 2025

CITI Program: IRB Training for Students and Faculty 2025